Bringing attention to e-cigarette pH as an important element for research and regulation

There is a lack of consensus among health experts and the tobacco control community regarding the potential population impact of the rapidly spreading use of electronic cigarettes, or e-cigarettes.[1–3] Dual use of e-cigarettes with regular cigarettes and initiation in non-tobacco users are among common concerns. Some experts, however, believe that complete switching to e-cigarettes could potentially lead to significant reductions in smoking-induced morbidity and mortality among current smokers who experience difficulty giving up smoking. Indeed, the e-cigarette aerosol, which primarily consists of nicotine, humectants and flavouring agents, contains significantly lower levels of many toxic and carcinogenic chemicals that are abundant in the smoke of regular cigarettes.[4]

The common toxicity concerns regarding e-cigarette use include nicotine toxicity, the potential long-term effect of propylene glycol (humectant) inhalation, and the lack of quality control oversight over the production of some e-cigarettes and refill fluids.[1 2 5] The latter can result in inconsistencies of nicotine content and contamination with various toxic and carcinogenic agents, including tobacco-specific nitrosamines.[4] One important factor that apparently has not been considered so far is the pH of e-cigarette liquid and of the resulting aerosol. It is known that pH determines the fraction of total nicotine that is present in the biologically available unprotonated form. Furthermore, at concentrations present in e-cigarette cartridges (roughly 6–24 mg/mL), the pH of unbuffered aqueous solutions of nicotine would be in the range 10.3–10.6, which is physiologically incompatible with the normal cellular function.

METHODS

We analysed the pH of several e-cigarette brands that offer disposable replacement cartridges of different nicotine 'strength'. The replacement cartridges were purchased online, via websites of the corresponding manufacturers. We included traditional/classic and menthol flavors. To measure pH, the contents of each cartridge were removed, extracted with 10 mL ultrapure water, and the pH of the aqueous extracts was measured with a pH meter according to our standard protocol.[6]

Table 1 pH of electronic cigarette replacement cartridge liquids

E-cigarette brand/nicotine content*	pH† Regular	pH† Menthol
Green smoke (%)	*Absolute tobacco*	*Menthol ice*
0.0	6.74±0.08	7.70±0.03
0.6	9.04±0.04	9.17±0.03
1.2	9.17±0.02	9.46±0.03
1.8	9.31±0.03	9.56±0.06
2.4	9.37±0.04	9.60±0.06
NJOY (%)	*Traditional*	*Menthol*
0.6	6.21±0.01	NA‡
1.2	7.47±0.03	8.16±0.03
1.8	7.91±0.05	8.20±0.06
V2 (%)	*Red*	*Menthol*
0.0	7.46±0.02	7.11±0.03
0.6	9.19±0.06	8.98±0.10
1.2	9.41±0.05	9.19±0.07
1.8	9.56±0.05	9.39±0.05
2.4	4.78±0.09	9.42±0.04
Blu (mg)	*Classic tobacco*	*Magnificent menthol*
0 g	5.48±0.03	6.11±0.02
8	7.36±0.08	8.50±0.03
12	7.62±0.02	8.45±0.01
16	7.97±0.02	8.48±0.01

*Nicotine content according to the label.
†Means and SDs of three measurements.
‡NA, not analysed. This variety was not available at the time of purchase.

RESULTS

The results (see table 1) demonstrate that the pH of e-cigarette cartridge content ranges widely, from 4.78 to 9.60, depending on the brand and nicotine level. While the pH of nicotine-free cartridges is generally neutral or even slightly acidic, over 50% of nicotine-containing cartridges have a pH greater than 9. Within individual brands, the pH generally increases with increasing nicotine content, and the pH of menthol-flavored varieties is generally higher than that of traditionally flavored ones. Only the V2 and Blu Magnificent Menthol brands did not follow one or both of these trends.

DISCUSSION

The variation of pH among e-cigarette brands that contain the same amount of nicotine, and even among different flavors within the same brand, is an important finding. Since pH determines the amount of nicotine that is present in the biologically available unprotonated form, e-cigarettes with the same nicotine content, but different pH, may deliver different doses of nicotine to users. Thus, potential pH variations represent an important factor that should be taken into account while designing and interpreting research studies involving e-cigarettes. This is also important for potential regulatory considerations.

Another significant observation is that most of the tested e-cigarette brands have basic pH. It is important to acknowledge that long-term and potentially life-long e-cigarette users are likely to emerge. Indeed, it has been reported that many long-term former smokers who are using e-cigarettes smoke them on a daily basis.[7 8] The long-term effect of chronic aerodigestive tract exposure to e-cigarette aerosol that has high pH is not known, and potentially represents an unrecognised health concern related to long-term e-cigarette use.

In summary, this report brings attention to an apparently unrecognised important issue that has to be addressed in e-cigarette research—variable and mostly

What this paper adds

This study highlights an apparently unrecognized important issue that has to be taken into account in e-cigarette research – variable and mostly highly basic pH of e-cigarette fluid.

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highly basic pH of e-cigarette aerosol. The public health community involved in e-cigarette research, as well as regulatory agencies, should take note of this issue.

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